1.

(a) (Theory)

Noise factor is defined as the total output (or input) noise power of the system divided by the output (or input) noise power due to the source alone.

$$\text{Noise factor } F = \frac{Psys + Pni}{Pni}$$

$$SNR_{in} = \frac{Psig}{Pni} \qquad SNR_{out} = \frac{Psig}{Pni + Psys}$$

$$\frac{SNR_{in}}{SNR_{out}} = \frac{Pni + Psys}{Pni} = \text{noise factor } F$$

(b) (Theory)

$$\frac{SNR_{in}}{SNR_{out}} = \frac{Pni + Psys}{Pni} = \text{noise factor } F$$

$$NF\ (dB) = SNR_{in}\ (dB) - SNR_{out}\ (dB)$$

We require a certain $SNR_{out} = SNR_{det}$ at the detector in order to recover the signal. Thus:

$$SNR_{in}\ (dB) = SNR_{det}\ (dB) + NF$$

$$SNR_{in}\ (dB) = Psig\ (dB) - Pni\ (dB)$$

We define the sensitivity of the system as the minimum input signal that we can successfully detect:

$$\text{Sensitivity} = Psig\ (dB) = SNR_{in}\ (dB) + Pni\ (dB)$$

$$= Pni\ (dB) + NF + SNR_{det}\ (dB)$$

Pni = input (received) noise.

(c) (New computed example)

$$F = \frac{vnt^2}{G1G2G3G4vns^2} = F1 + (F2\text{-}1)/G1 + (F3\text{-}1)/G1G2 + (F4\text{-}1)/G1G2G3$$

F=14.409

ANSWERS

(d) (Application of Theory)

The one with the highest gains first, since the noise factor is given by:

$$F = \frac{vnt^2}{G1G2G3G4vns^2} = F1 + (F2-1)/G1 + (F3-1)/G1G2 + (F4-1)/G1G2G3$$

Hence subsequent blocks have their noise contribution attenuated by the previous gains.

(e) (Application of Theory)

No. There would be aliasing because the sampling frequency is lower than twice the bandwidth of the signal.

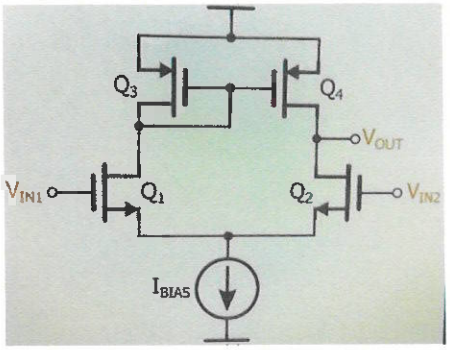
(f) (Application of Theory)

This is the second harmonic of the local oscillator (the frequency of the latter can be calculated from the frequency of the intermediate frequency).

ANSWERS

2.

(a) (Theory)



(b) (Application of theory)

The solution is a drawing showing sources at the input of the transistors. The noise corresponding to the bias current is not important because it is common mode out. The others are significant because two of them appear at the inputs (and since they are uncorrelated, get added together), and the other two are in a low impedance node with respect to the input.

(c) (New computed example)

$v_{eq}^2 = (2v_{n1}^2 + 2v_{n3}^2(gm3/gm1)^2)\,[gm1/(gds2+gds4)]^2$

(d) (Application of theory)

The solution is a drawing showing two resistors, going from the sources of Q1 and Q2 to the common mode bias node respectively.

(e) (New computed example)

$v_{eq}^2 = 2v_{n1}^2 + 2v_{n3}^2(gm3/gm1)^2 + 8KTR$

(f) (Application of theory)

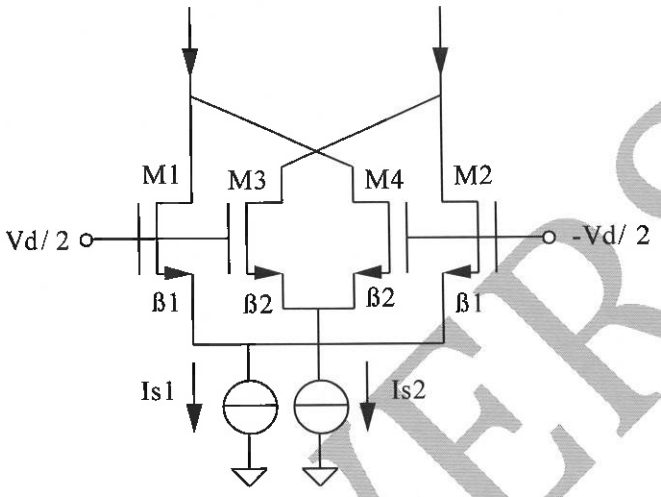
In order to achieve an increased linearity the resistors should be large, but larger resistors also increase noise.

ANSWERS

3.

(a) (Theory)

One transistor is, as the name says, just one transistor. Two would be a differential pair (as for example in Question 2(a)). For four there could be more than one option. For example:



(b) (Application of theory)

This question has different answers depending on the topologies chosen. In general, choosing differential topologies helps with linearity because even order harmonic are mathematically eliminated. Larger number of transistors, also allow further harmonics cancellation. For example, in the case drawn in the example, it is possible to choose the transistors sizes so that the third order harmonic is eliminated.

(c) (Application of theory)

Matching is important because otherwise even order harmonics will not cancel out.

(d) (Application of theory)

The answer to this question depends on the topologies chosen in (b). In the example, the differential pair and the four transistor topologies could be mixers if one of inputs is taken from the source. Exactly how would depend on how that source is designed. In the simplest case of a single transistor, the second input to the mixer would be through the gate of that transistor.

(e) (Application of theory)

This will depend on the chosen topology. For the one in the example:

$Gm = (Is\beta 1)^{1/2}-(Is\beta 2)^{1/2}$
$Iout=GmVd$

ANSWERS

(f) (New Theory)

No. Transistors in weak inversion cannot operate at such high frequencies.

4.

(a) (Theory)

Because of the frequency range Flicker noise is negligible. The noise is thermal and is given by:

$vn^2 = 8kT/3gm\ (V^2/Hz)$

(b) (New computed example)

The solution would be a sketch with a low pass filter type of response. In that frequency range it would look like a monotonically decreasing line with a slope of -20dB/dec (if representing power and dB defined as 10log)

(c) (Application of theory)

No. The differential pair has got a very small linear range, and 0.3V is large signal.

(d) (Application of theory)

This question has many possible answers. In general the student should show the topology of a linearized transconductor.

(e) (New theory)

This answer would depend on what the student answered for (d). For a differential topology, the solution would be IP3, because that would be the dominant harmonic, considering properly matched differential topologies get rid of even order harmonics.

(f) (Application of Theory)

It is a fully differential integrator.

Advantages of crossed-couple: no need of fully balanced input, fully linear in absence of mismatch, programmability, no common mode dependency, resilient to threshold voltage variations.

ANSWERS